



Institute of Science and Technology (IST)

Department of CSE
1st year 2nd Semester
Preparation Test 2023

Course Title: Statistics and Probability (CSE-510227)

Total Marks: 80, Time: 3 Hours

[Answer any four questions]

- 1a) Write different sources of data used in statistics? 3
b) Write the different steps of a questionnaire. 2
c) Why needed graphical representation of data? What are various types of charts and diagrams used in presenting statistical data? Explain one of them. 5
d) The price of (thus tk) 40 computers are given below: 10
20 30 45 55 36 56 62 18 20 65 35 69 22 39 49 50 53
67 65 58 37 18 65 48 16 45 19 53 23 58 30 49 45 56 45
i) Construct a frequency distribution from the above data.
ii) Construct a stem and leaf diagram for this data

2a) What are the essential characteristics of an ideal average? Which measure of central tendency is best? 3

b) For two observations x & y prove that $A.M. \geq G.M. \geq H.M.$ 4

c) The following table gives the daily wage distribution of the employees of a commercial organization:

Profits per shop	250-300	300-350	350-400	400-450	450-500	500-550
No. of shops	20	10	11	25	14	30

- i) Compute median and mode. 10
ii) Draw a histogram and then find modal value?
d) A car travels 25 miles at 25 mph, 25 miles at 50 mph, 25 miles at 75 mph. Find average speed? 3

3a) What is coefficient of variation? What purpose does it serve? Also distinguish between standard deviation and coefficient of variation. 5

b) If $\bar{x}$ and s denote respectively mean and standard deviation of a set of non-negative quantities, prove that $\bar{x}\sqrt{n-1} \geq s$. 5

c) The temperature of cities A and B in a winter season are given below 10

Temperature of city A (in degree Celsius) 18 20 22 24 26

Temperature of city B (in degree Celsius) 11 14 15 17 18

Find which city is more consistent in temperature changes?

4a) What is rank correlation? Formulate rank correlation formula. 6

b) Prove that correlation coefficient is the geometric mean of two regression coefficients. 4

c) Find coefficient of correlation and correlation ratio price and sales from the following data: 10

Price (in taka)	103	98	85	92	90	84	88	90	93	95
Sales (units)	500	610	700	630	670	800	800	750	700	680

- 5a) What is regression? How you estimate regression parameters using least square method? 6
 b) The following information : 10

	Advertisement expenditure (Tk lakhs)(x)	Sales (Tk crores)(y)
Average	45	98
standard deviation	12	23

Coefficient of correlation $r=0.91$.

i) Find the regression equations x on y and y on x . ii) Find probable sales when advertisement cost is 50 lakhs.

c) Display the data in a box plot, 12,5,6,12,6,14,16 .

4

6a) What is association of attributes? Write different method of association of attribute. When we use contingency table? 6

b) How you measure of association for nominal data? 4

c) In an examination, at which 600 candidates appeared, boys outnumbered girls by 16% of all candidates. Number of passed exceeded the number of failed candidates by 310. Boys failing in the examination numbered 88. Find the Yule's coefficient of association between male and success in examination. 10